EDAV - Assessed Coursework 2

1 Data Cleaning and Preprocessing

Missing temperature, rain, and wind speed values were interpolated, with rainfall smoothed by a rolling 7-day average. No duplicates were found. We applied a log transformation to normalize numeric variables before outlier inspection via box plots. Potential outliers appeared in average daily price, wind speed, and temperature. Using Z-score detection, we Winsorized outliers at the 1st and 99th percentiles. Finally, the dataset was reviewed for completeness.

2 Univariate Analysis

For our univariate analysis, we chose **National Energy Demand**. The end goal of the project for which we are performing EDA is forecasting energy demand and achieving net zero by 2050. In this context, a univariate analysis on National Energy Demand is the most appropriate, laying the foundation for prediction and time-series modeling for this variable later on in the project, noting the seasonality.

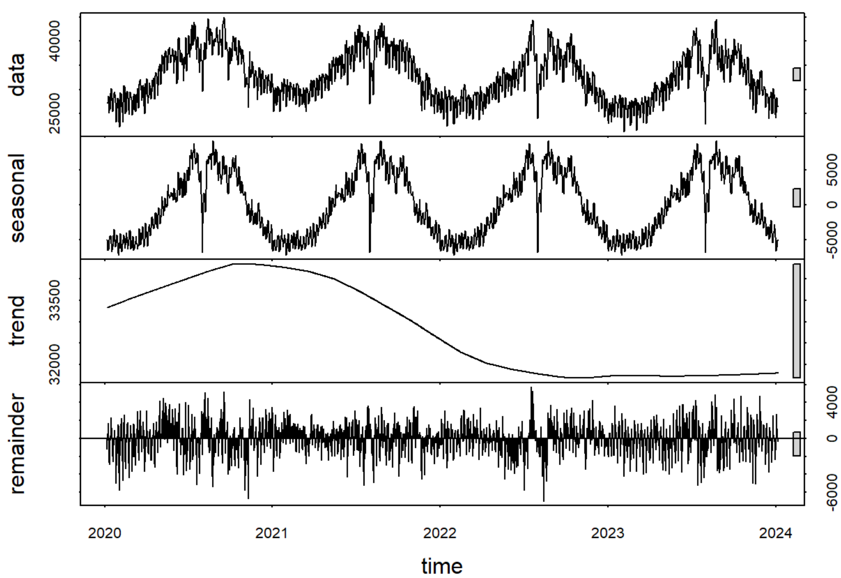

Figure 1: Time series decomposition of National Demand

Time series decomposition was carried out to analyse **seasonal**, **trend** and **residual** components. There is a clear **seasonal component** in the variable, exhibiting an annual cycle with peaks in winter months and troughs in summer months. This increased demand in winters can be attributed to increased heating and lighting needs during colder and darker months. Further analysis shows dips around Christmas Day and New Year, possibly due to business closures during holiday period. The **Trend component** shows a slight increase between 2020 and 2021, possibly explained by post-COVID-19 re-openings, which is followed by a gradual decrease in the national demand, pointing towards improved overall system efficiencies. However, the **fluctuation scale** across trend component is insignificant compared to seasonal component. Removing the seasonal and trend components, the **residuals exhibit no strong autocorrelation and resemble a weakly stationary process**, with some extreme spikes suggesting unexpected variations, possibly due to power outages or anomalous weather conditions.

The **dominant frequency** f_{max} in the dataset was identified as **0.00267** through spectral analysis. This corresponds to a **periodicity** of **375 days**, approximating a yearly cycle in national demand.

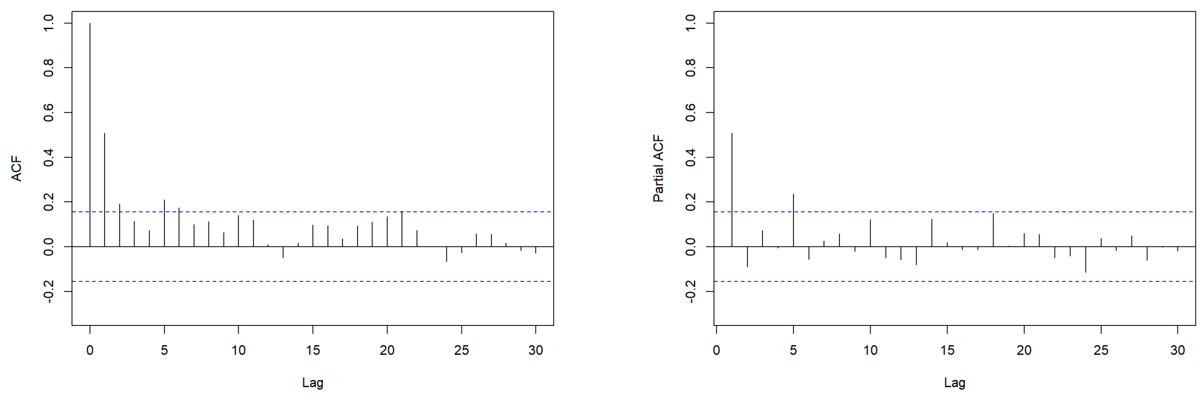

Figure 2: Seasonally differenced ACF (left) and PACF (right) plots of weekly aggregated National Demand

To achieve stationarity, seasonal differencing with a lag of 365 days was conducted on daily national demand, which is given as $\hat{Y}_t = Y_t - Y_{t\_365}$. Seasonal differencing on daily data points retained some seasonal component as it showed spikes around every 7^{th} lag. A second approach was taken to apply seasonal differencing with a **lag of 52 weeks** on **weekly aggregated national demand**. The Autocorrelation Function (ACF) plot shows significant correlation at lag 1, which rapidly decays over the next 5 lags. The subsequent lags fall within the 95% confidence interval, indicated by blue dashed lines. Analysing the Partial Autocorrelation Function (PACF) plot suggests that the data follows AR(1) process, as PACF cuts off at lag 1; however, the spike at lag 5 is noted. Thus, it is suggested to utilise SARIMA model for modelling this data. To confirm the stationarity, Augmented Dickey-Fuller (ADF) test was conducted that yielded a p-value of **0.01** with **lag order 5**, implying stationarity.

3 Multivariate Analysis

3.1 Motivation for variable selection : Causal and Pairwise Relationships

In this section, we examine the relationship between solar energy (used as a proxy for sunlight energy received by earth) and energy demand. The theoretical causal chain suggests that in winter, lower solar intensity reduces temperatures. Simultaneously, colder temperatures increase heating demand while solar generation remains low, exacerbating energy shortages. The opposite effect occurs in summer. Based on this, visualisations have been segmented into April–September and October–March, further divided into weekdays and weekends, to highlight intra-week cyclic variations in correlations likely due to household energy usage patterns. The minimum temperature variable was selected as it was found to be highly correlated with maximum temperature, but the daily average of the two variables was also considered.

The Granger causality test evaluates whether past values of one time series enhance the prediction of another, thereby establishing a directional relationship. Using the Schwarz Criterion to guide lag selection, we tested whether solar generation predicts energy demand and temperature. The results indicate solar energy Granger-causes both temperature and energy demand, and temperature also Granger-causes demand. An illustration of the causal chain can be found in Figure 3 (Left), where the p-values within blocks are associated with the receiver variable based on the previous link in the chain.

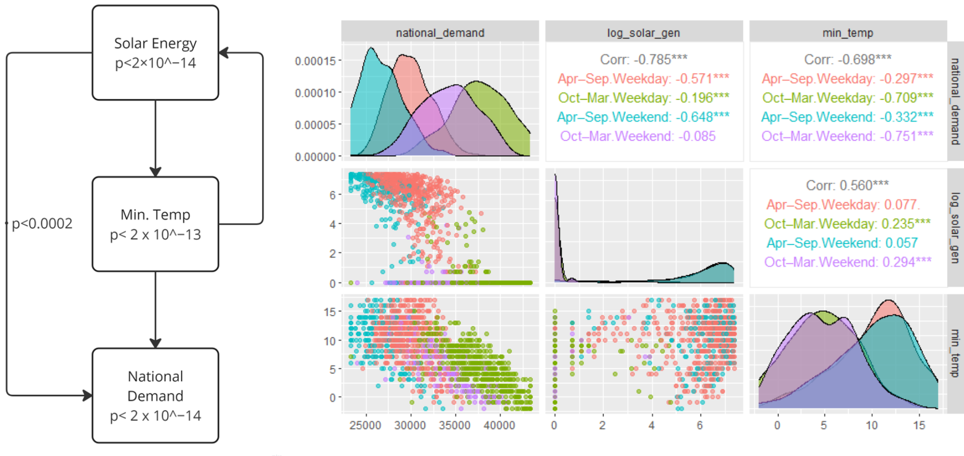

Figure 3: (Left) Casual Chain of Solar Energy, Temperature and National Demand with p-values; (Right) Pairwise scatterplots and a correlation matrix for national demand, log-transformed solar generation (using log1p(x) due to zero values), and minimum temperature, with density curves along the diagonals.

The log transformation in Figure 3 (right) helps visualise extreme outliers in solar generation. We use Spearman's correlation (method = "spearman") for its robustness to outliers and nonlinearity. Although national demand and log(solar generation) show a strong negative correlation, solar remains heavily skewed, which may inflate even Spearman's estimates. Splitting by season reveals Apr–Sep with a higher negative correlation—especially on weekdays when demand is higher(given a period)—while Oct–Mar shows a weaker or "inverted" intra-week relationship.

National Demand vs Min Temp exhibits consistent high negative correlations in Oct-Mar due to the increased demands in colder periods, the correlation is still negative but lower magnitude in warmer periods.

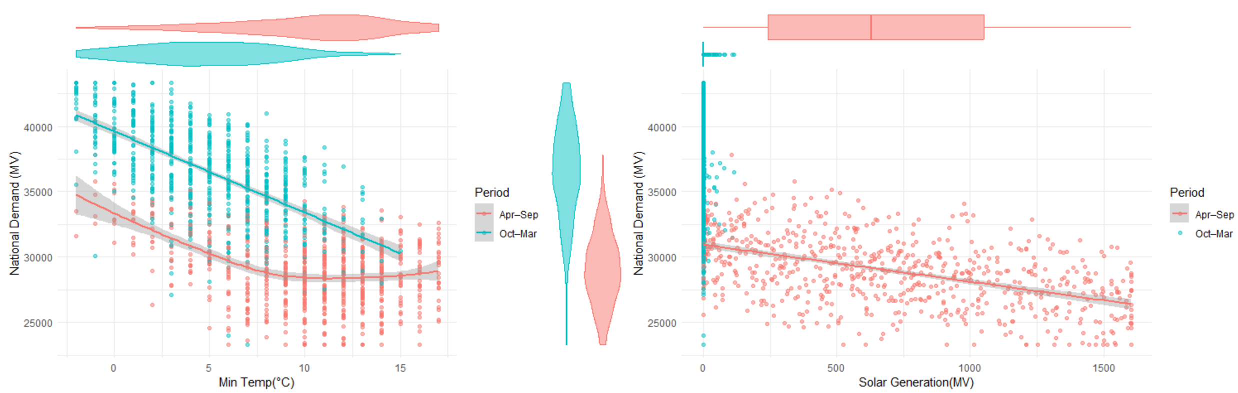

Figure 4: Scatterplots conditioned on seasonal period for National Demand vs Min Temp (Left) and Solar Generation (Right).

Figure 4, separated from the earlier pairwise plots, highlights seasonal effects more clearly. The scatter plots on the left show National Demand vs. Min Temp for both periods season: a linear trend (method = "lm") is used for Oct-Mar with default parameters and 0.95 confidence interval, while Apr–Sep specifically uses a loess fit (default parameters, 0.95 confidence, degree 2 polynomial). The trend would appear to be linear when without conditioning on Period, but the plot illustrates doing so exhibits alternative correlations. The wider intervals at lower temperatures reflect fewer observations and greater uncertainty. The violin plots give an indication of the National Demand and Min Temp distributions, conditioned on the seasonal period; the National Demand is positioned centrally to bridge between the two scatterplots, emphasising the density of zero or near-zero plots for Solar Generation in Oct-Mar, as well as the increasing generation in Apr-Sep.

On the right plot in Figure 4, Solar Generation is shown by a boxplot for Apr–Sep as it gives a clearer overview of the distribution in this period compared to a low density and narrow violin plot; this was explored initially. The boxplot shows distribution is slightly skewed towards lower values as well as the quartile positions. Meanwhile, Oct–Mar has many near-zero points and no obvious trend, although there are few outliers above zero which are likely the tail ends of the Oct-Mar period. In contrast, Apr–Sep data follows a more linear pattern with narrower confidence intervals (method = ”lm”, default parameters, 0.95 confidence). Observing this non-logarithmic version of the scatterplot in contrast to the version in Figure 3 gives a more interpretable view with real MV values alongside the violin/box plots, and focuses on the seasonal differences, excluding Weekday and Weekend.

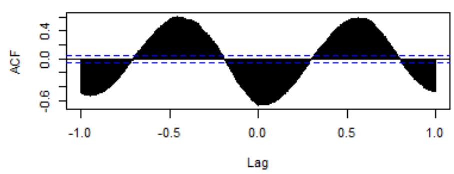

Figure 5: CCF: National Demand vs Solar Generation

The Cross-Correlation Function (CCF) measures the relationship between energy demand and solar generation at different time lags, capturing how their fluctuations align over time. The CCF plot (Figure 5), using a 365-day frequency, reveals a strong seasonal pattern, with alternating positive and negative correlations indicating that solar generation and energy demand peak at different times which is consistent with the time series plot on the left. While the CCF alone does not establish causality or confirm that solar generation leads demand, it highlights a consistent lead-lag structure. Given that previous Granger causality tests indicating that solar generation (used as a proxy for overall sunlight energy received) Granger-causes energy demand, this CCF pattern suggests that, if the causal link holds, solar generation can be used as a predictive factor for forecasting future energy demand. In addition, forecasting demand could also be explored with temperature.